I will keep the current six million tile corpus fixed and redesign only the model, training protocol, and evaluation so the paper centers on overlap-aware representation learning rather than new preprocessing.

Thought for 1m 14s >

MississippiFM

Proposed title

MississippiFM: Overlap-Aware Foundation Pretraining for Sub-Meter Remote Sensing

This title is strong. It identifies the model, the methodological contribution, and the spatial-resolution domain.

I would keep “remote sensing” rather than “aerial imagery” because the downstream benchmark will cover several remote-sensing tasks. The paper must remain clear that MississippiFM was pretrained on sub-meter aerial imagery, not on multimodal satellite data.

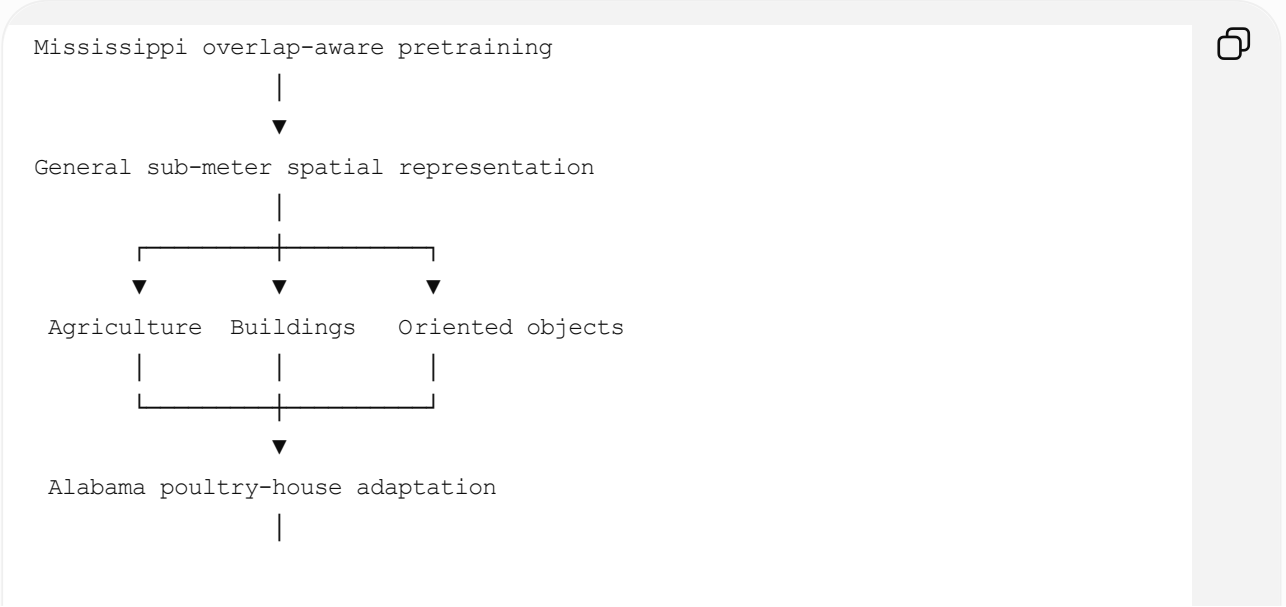
---

1. Refined paper thesis

> Can known geographic overlap between adjacent sub-meter aerial tiles provide dense self-supervision that improves transferable remote-sensing representations?

The paper should not focus only on whether Mississippi pretraining improves Alabama poultry-house detection.

The stronger argument is:



```
▼
Georgia geographic transfer
```

Poultry-house mapping becomes the most detailed application, while public benchmarks establish that MississippiFM learned broader remote-sensing features.

---

2. Final contribution package

Contribution 1. Geographically aligned self-supervised learning

Refined statement

> We introduce overlap-aware geographic alignment for self-supervised pretraining. Known overlap between adjacent NAIP tiles provides exact ground correspondences, allowing MississippiFM to align patch representations of the same geographic region across different crop positions and augmentations.

The important phrase is **exact ground correspondence**.

NeighborMAE already uses neighboring and overlapping Earth-observation images during masked pretraining. MississippiFM therefore cannot claim novelty simply from training with adjacent tiles. It must distinguish itself through explicit patch-to-ground coordinate matching and dense feature alignment. arXiv

What makes it distinct

```
Neighboring tiles only
Weak spatial relationship

Overlapping tiles
Known shared area

MississippiFM
Exact matching between corresponding ground patches
```

For two tiles, A and B:

1. Determine their known pixel offset.
2. Identify their shared geographic area.
3. Locate corresponding patch centers.
4. Encode both tiles with one shared ViT.
5. Align embeddings only for geographically corresponding patches.
6. Ignore nonoverlapping patches in the alignment loss.

This makes the overlap a source of dense supervision rather than duplicated training data.

Contribution 2. Sub-meter aerial foundation model

Refined statement

> We pretrain a Vision Transformer on more than six million 512 by 512 aerial tiles covering Mississippi at approximately 60-centimeter ground sampling distance, producing a foundation encoder specialized for fine-scale land structure, buildings, vegetation, roads, agricultural fields, and rural settlement patterns.

Do not write that six million tiles are six million independent geographic scenes. The tiles overlap by 50 percent.

Report:

- Total saved tiles.
- Approximate unique ground coverage.
- Source raster count.
- Counties represented.
- Acquisition years.
- Pretraining steps.
- Total tokens processed.

The project already has MIM, SparK, ViT, YOLO11, YOLO26, geospatial inference, and downstream segmentation infrastructure. Readme 03_Machine_Learning_and_SSL

Contribution 3. Strict geographic and scale-shift benchmark

Refined statement

> We introduce a leakage-resistant transfer benchmark spanning states, counties, acquisition years, ground sampling distances, datasets, and downstream tasks. The benchmark measures frozen representation quality, label efficiency, full fine-tuning, and geographic transfer from Mississippi to Alabama and Georgia.

I would add “scale-shift” because your evaluation will include different GSDs.

Benchmark axes

Axis	Evaluation
State	Mississippi, Alabama, Georgia
County	Held-out counties

Axis	Evaluation
Time	Different acquisition years
Resolution	Different GSDs
Dataset	Internal and public
Task	Segmentation, detection, regression
Labels	Frozen, few-shot, full
Context	Local and large-area

---

3. Freeze the current data

No image reprocessing should occur during the next stage.

Keep unchanged

- Existing six million JPG tiles.
- Existing 512 by 512 resolution.
- Existing 50 percent overlap.
- Existing Mississippi source imagery.
- Existing Alabama annotations.
- Existing Georgia imagery.
- Existing GeoTIFF-to-JPG conversion.
- Existing trainable data loader.

Do not repeat

- GeoTIFF slicing.
- JPG regeneration.
- Resolution conversion.
- New statewide data collection.
- Full corpus deduplication.
- Reorganization of all six million files.
- Multispectral conversion.

Only required metadata work

Create one lightweight adjacency index.

```
tile_a
tile_b
source_raster
row_offset
column_offset
overlap_x1
overlap_y1
overlap_x2
overlap_y2
```

This does not modify the images. It only tells the training loader which existing tiles overlap.

If filenames already contain the source and window coordinates, generate the index from those names. If the coordinates exist in previous slicing logs, read those logs. Do not reopen or reslice the original GeoTIFFs.

This adjacency index is essential. Without known tile offsets, the method becomes neighbor-based SSL rather than geographically aligned SSL.

---

4. Final MississippiFM model

Main backbone

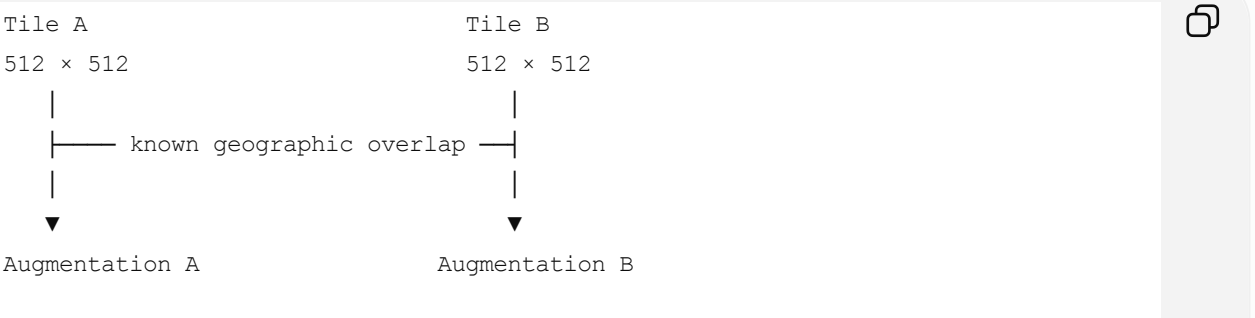
DINOv3 ViT-B/16

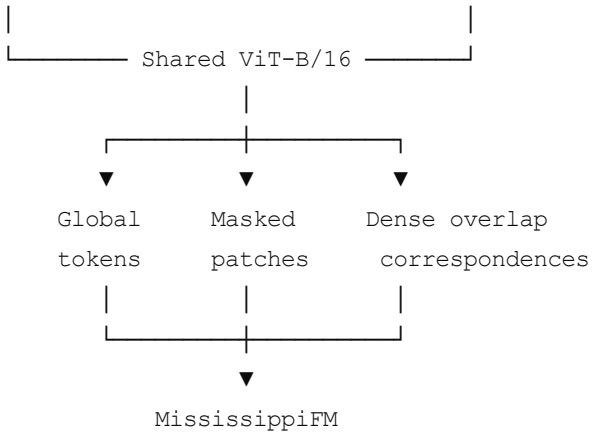
Use:

- ViT-S/16 for pilot experiments.
- ViT-B/16 for the main paper.
- ViT-L/16 only as an optional scaling experiment.

DINOv3 is a strong starting point because it provides global and dense self-supervised features and introduces Gram anchoring to protect dense spatial representations during long training schedules. arXiv +1

Main architecture





---

5. Training objectives

Use four complementary objectives.

$$L_{\text{total}} = L_{\text{global}} + \lambda_p L_{\text{patch}} + \lambda_o L_{\text{overlap}} + \lambda_g L_{\text{gram}}$$

5.1 Global self-distillation

A DINO-style global objective aligns semantic representations across different views.

It should capture:

- Rural versus settled areas.
- Forest and agricultural context.
- Road networks.
- Farm layouts.
- Large building arrangements.
- Broad land-use structure.

5.2 Masked patch prediction

Use an iBOT-style masked patch objective.

It should capture:

- Roof texture.
- Field boundaries.
- Vegetation patterns.
- Building edges.
- Roads.
- Small structures.
- Fine ground texture.

5.3 Geographic overlap alignment

This is the central contribution.

For corresponding patch sets P_A and P_B:

$$L_{\text{overlap}} = \frac{1}{N} \sum_{i=1}^{N} \left(1 - \cos\left(z_i^A, z_i^B\right)\right)$$

Where:

- z_i^A is a patch embedding from tile A.
- z_i^B represents the same ground location in tile B.
- N is the number of valid geographic correspondences.

Use a small projection head for this loss. Remove it after pretraining.

5.4 Gram anchoring

Use the original DINOv3 encoder as a frozen reference.

The anchor prevents domain pretraining from damaging useful dense spatial relationships. DINOv3 introduced Gram anchoring specifically to address dense-feature degradation during extended training. arXiv +1

---

6. Recommended initial loss weights

Objective	Weight
Global distillation	1.0
Masked patch	1.0
Geographic overlap	0.5
Gram anchor	0.1

Test geographic-overlap weights:

```
0.1
0.25
0.5
1.0
```

Use ViT-S for this ablation. Do not repeat every weight experiment with ViT-B.

7. Training pair strategy

Each batch should contain both overlapping and independent samples.

Pair type	Share
Overlapping adjacent tiles	50 percent
Two views of one tile	25 percent
Independent tiles	25 percent

Why include independent tiles

If every batch contains only adjacent tiles, the model may over-specialize to local correspondence.

Independent tiles preserve:

- Statewide visual diversity.
- Global semantic learning.
- Different soil and vegetation conditions.
- Different rural and urban contexts.

---

8. Resolution and context strategy

Training ViT-B directly at 512 by 512 for the entire campaign will be expensive because attention cost grows rapidly with token count.

Use two stages.

Stage 1. Main pretraining

Setting	Value
Input view	256 or 320
Backbone	ViT-B/16
Global views	2
Local views	4 to 6
Precision	BF16

Setting	Value
Objective	Full MississippiFM
Data	Existing six million tiles

Random crops must retain the transformation coordinates so geographic patch matches remain correct.

Stage 2. High-resolution adaptation

Setting	Value
Input	512
Duration	Last 10 to 15 percent
Learning rate	Reduced
Objectives	Patch, overlap, Gram
Global crop	Full tile

This allows MississippiFM to learn full-tile spatial context without paying 512-resolution training cost for the entire run.

---

9. Practical training funnel

```
ViT-S pilot
100,000 to 500,000 existing tiles
            |
            ▼
Objective and loss selection
            |
            ▼
ViT-S full-corpus confirmation
            |
            ▼
ViT-B full Mississippi training
            |
            ▼
512-resolution adaptation
            |
            ▼
Frozen benchmark
```

```
                │
                ▼
Downstream fine-tuning
```

Pilot experiments

Run these first:

ID	Method
P0	DINOv3 unchanged
P1	Mississippi MIM
P2	DINO plus iBOT
P3	DINO plus iBOT plus overlap
P4	P3 plus Gram anchoring

P4 is the proposed MississippiFM.

---

10. Refined downstream evaluation

The proposed concepts are valuable, but they are not all standard downstream tasks.

Organize them into two levels:

1. Standard benchmark tasks.
2. Spatial-representation diagnostic probes.

---

11. Standard benchmark tasks

11.1 Agricultural land patterns

Task

Semantic segmentation of agricultural patterns.

Dataset

Agriculture-Vision.

Agriculture-Vision was introduced for semantic segmentation of large aerial farmland imagery and contains 94,986 aerial images with agricultural pattern labels.
CVF Open Acc… +1

What it tests

- Field structure.
- Waterways.
- Agricultural anomalies.
- Vegetation patterns.
- Boundary understanding.

Metrics

- mIoU.
- Macro Dice.
- Boundary F1.
- Per-class IoU.

11.2 Settlement context and land cover

Task

High-resolution land-cover segmentation.

Dataset

OpenEarthMap.

OpenEarthMap includes 5,000 high-resolution images, 2.2 million annotated segments, 97 regions, and eight land-cover classes at approximately 0.25 to 0.5 m GSD.
CVF Open Acc… +1

What it tests

- Settlement context.
- Buildings.
- Roads.
- Vegetation.
- Agricultural land.
- Water.
- Bare land.

Metrics

- mIoU.

- Building IoU.
- Road IoU.
- Vegetation IoU.
- Worst-region mIoU.

11.3 Roof and building structure

Task

Building-footprint segmentation.

Dataset

Inria Aerial Image Labeling.

What it tests

- Roof texture.
- Building boundaries.
- Dense settlements.
- Sparse rural buildings.
- Geographic transfer.

Metrics

- Building IoU.
- Dice.
- Boundary F1.
- Building-count error.
- Area error.

11.4 Oriented structures

Task

Oriented object detection.

Dataset

DOTA.

DOTA contains large aerial images from several platforms, with objects spanning different orientations, shapes, and scales. CVF Open Acc… +1

What it tests

- Orientation awareness.
- Long rectangular structures.
- Small-object localization.
- Rotation robustness.
- Dense spatial representations.

Metrics

- OBB AP50:95.
- AP50.
- AP75.
- Per-class AP.

11.5 Poultry-house mapping

Alabama

- Supervised fine-tuning.
- In-state held-out evaluation.
- Label-efficiency evaluation.
- YOLO and ViT-head comparisons.

Georgia

- Zero-shot state transfer.
- Threshold-free transfer using Alabama validation settings.
- Optional few-shot adaptation.
- County-level analysis.

Primary tasks

- Instance segmentation.
- Oriented detection.
- House-level geospatial matching.
- Farm-level clustering.

Your existing pipeline already supports segmentation, tiled inference, mask suppression, farm clustering, GIS export, and county-level analysis. Readme 05_GIS_PostProcessing_and_Analy…

---

12. Spatial-representation diagnostic probes

These probes directly evaluate the concepts you listed.

Probe 1. Agricultural land-pattern recognition

Predict:

- Row structure.
- Field boundary.
- Waterway.
- Bare patch.
- Vegetation anomaly.

Use frozen MississippiFM features and a linear segmentation head.

Probe 2. Settlement-context classification

Classify each tile into:

- Isolated rural structure.
- Farm complex.
- Low-density settlement.
- High-density settlement.
- Industrial or commercial.
- Mostly undeveloped.

Report macro F1.

Probe 3. Road-building relationship

Predict:

- Nearest-road distance category.
- Building connected to road.
- Building isolated from road.
- Road-adjacent building density.

This can use road and building masks from a public dataset. No Mississippi relabeling is required.

Probe 4. Field structure

Predict:

- Field-boundary mask.
- Dominant field orientation.
- Field-size category.
- Regular versus irregular field layout.

Probe 5. Building density

Regress:

- Building count.
- Building area percentage.
- Buildings per square kilometer.
- Mean nearest-neighbor distance.

Metrics:

- MAE.
- RMSE.
- R^2.

Probe 6. Roof and vegetation texture

Evaluate patch retrieval.

Example query:

```
Metal roof patch
        |
        ▼
Retrieve similar roof patches
and reject vegetation or roads
```

Metrics:

- Recall at 1.
- Recall at 5.
- mAP at 10.

Probe 7. Large-scale spatial context

Compare the same encoder using:

- Local crop only.
- Full 512 tile.
- Tile plus one adjacent tile.
- Tile plus four-neighbor context.

Measure the improvement for:

- Poultry detection.
- Settlement classification.
- Building-density regression.
- Agricultural segmentation.

This probe directly tests whether MississippiFM learned more than local texture.

---

13. Context should be part of the model evaluation

Your seven concepts can be grouped into three representation scales.

Scale	Concepts
Local	Roof and vegetation texture
Object	Buildings, roads, field boundaries
Scene	Settlement, density, land patterns
Regional	Large-scale spatial context

The paper should show that overlap-aware training helps more at the object and scene levels than ordinary single-tile SSL.

That is a stronger scientific story than simply showing higher poultry-house mAP.

---

14. Baseline models

Foundation-model track

Use one common downstream head.

Baseline	Purpose
Random ViT-B	No pretraining
ImageNet ViT-B	Supervised natural images
MAE ViT-B	Generic masked modeling
DINOv2 ViT-B	Generic SSL
DINOv3 ViT-B	Strong dense SSL
Scale-MAE	Scale-aware RS baseline
NeighborMAE	Neighbor-overlap baseline
Mississippi MIM	Domain-only baseline
MississippiFM	Proposed

Scale-MAE explicitly models known geographic scale and reconstructs low- and high-frequency content. CVF Open Acc… +1

Applied poultry-mapping track

Model	Initialization
YOLO11	COCO
YOLO11	Mississippi SparK
YOLO26	COCO
YOLO26	Mississippi SparK
ViT-B decoder	DINOv3
ViT-B decoder	MississippiFM

Keep these results separate from the common-head foundation-model comparison.

---

15. Downstream adaptation modes

Run every ViT backbone under three protocols.

Frozen

- Frozen backbone.
- Linear classification head.
- Linear dense head.
- No encoder updates.

This provides the cleanest representation comparison.

Parameter-efficient

- LoRA rank 8.
- Train task head.
- Less than 5 percent trainable parameters.

Full fine-tuning

- Entire encoder trainable.
- Same head and training schedule for all models.

A model that improves only under full fine-tuning has a weaker foundation-model claim than one that also improves frozen transfer.

16. Alabama and Georgia evaluation

Alabama split

Use the existing dataset.

Do not regenerate the images.

At minimum, ensure that all images from one farm remain in one split.

```
AL-Train
Fine-tuning

AL-Val
Hyperparameters and thresholds

AL-Test
In-state generalization
```

Georgia split

```
GA-Test
Locked zero-shot evaluation

GA-Adapt
Optional few-shot experiments
```

Georgia protocols

Protocol	Training
Zero-shot	Alabama only
Calibration	GA-Adapt thresholds only
Few-shot	1, 5, 10, 25 farms
Full adaptation	Complete GA-Adapt

Never tune on GA-Test.

17. Poultry-house evaluation levels

Tile level

- Box AP50:95.
- Mask AP50:95.
- OBB AP50:95.
- Boundary F1.

House level

- Precision.
- Recall.
- F1.
- Centroid error.
- Roof-area error.
- Duplicate rate.
- False positives per square kilometer.

Farm level

- Farm precision.
- Farm recall.
- Farm count MAE.
- House count error per farm.
- Farm-centroid error.

Geographic transfer

$$\text{State transfer retention} = \frac{\text{Georgia score}}{\text{Alabama test score}}$$

Report both retention and absolute performance loss.

---

18. Core ablation experiments

Overlap ablation

Experiment	Pairing
A0	No overlap pairs
A1	Adjacent tiles as global positives
A2	Shared-area reconstruction
A3	Exact patch correspondence

Experiment	Pairing
A4	A3 plus Gram anchoring

The key result is A3 versus A1 and A2.

Overlap amount

Use the existing 50 percent data.

Simulate smaller visible overlap during training:

- 10 percent.
- 25 percent.
- 50 percent.
- 75 percent of available shared area.

No new tiling is required.

Correspondence quality

Compare:

- Exact correspondence.
- Offset disturbed by one patch.
- Offset disturbed by two patches.
- Neighbor identity without correspondence.

This tests whether geographic precision actually matters.

Context ablation

- 256 crop.
- 320 crop.
- Full 512 tile.
- Tile plus neighbor context.

Data scale

Use subsets of the existing corpus:

- 100,000.
- 500,000.
- 1 million.
- 3 million.
- More than 6 million.

No data regeneration is necessary.

19. Checkpoint selection

Do not select the best model from pretraining loss.

Use four frozen validation tasks:

1. Agricultural pattern segmentation.
2. Building segmentation.
3. Poultry-house segmentation.
4. Overlap-patch correspondence.

Create a normalized validation score:

$$S = 0.25S_{\text{agriculture}} + 0.25S_{\text{building}} + 0.25S_{\text{poultry}} + 0.25S_{\text{correspondence}}$$

Use this score only on validation partitions.

---

20. Main paper narrative

Problem

Sub-meter aerial imagery contains strong spatial continuity, but common self-supervised methods treat overlapping tiles as unrelated images or duplicate samples.

Method

MississippiFM uses known tile overlap to align dense ViT patch representations corresponding to the same geographic ground area.

Scale

The model is pretrained on more than six million 60-centimeter Mississippi aerial tiles.

Evidence

The model is evaluated across:

- Mississippi, Alabama, and Georgia.
- Different counties.
- Different imagery years.
- Different GSDs.
- Agricultural segmentation.
- Building segmentation.
- Oriented detection.
- Poultry-house mapping.
- Frozen, LoRA, and full fine-tuning.

Impact

The method improves representation transfer for sub-meter remote sensing, with poultry-house mapping as a detailed agricultural application.

---

21. Final contribution wording for the paper

Contribution 1. Geographically aligned SSL

We introduce an overlap-aware self-supervised objective that uses known geographic correspondences between adjacent aerial tiles to align dense patch representations of the same ground region.

Contribution 2. Sub-meter aerial foundation model

We develop MississippiFM, a Vision Transformer pretrained on more than six million 60-centimeter aerial tiles, targeting fine-scale rural, agricultural, settlement, building, road, and vegetation structure.

Contribution 3. Strict geographic benchmark

We evaluate transfer across states, counties, acquisition years, GSDs, datasets, and downstream tasks using leakage-resistant splits, frozen probes, parameter-efficient tuning, full fine-tuning, and zero-shot geographic transfer.

Contribution 4. Spatial understanding analysis

We introduce diagnostic probes covering agricultural land patterns, settlement context, road-building relationships, field structure, building density, roof and vegetation texture, and large-scale spatial context.

I recommend including Contribution 4. It converts your proposed downstream themes into a clear scientific analysis rather than leaving them scattered across the experimental section.

---

22. Immediate model and training work

First

- Freeze the existing tiles.
- Create the adjacency index.
- Implement exact overlap patch matching.
- Add overlap loss to ViT-S.

- Use the current MIM as a baseline.
- Add DINOv3 initialization.
- Add iBOT patch prediction.
- Add Gram anchoring.

Second

Run a limited ViT-S study:

```
DINOv3
Mississippi MIM
DINO plus iBOT
DINO plus iBOT plus overlap
Full MississippiFM
```

Use 100,000 to 500,000 existing tiles.

Third

Evaluate frozen representations on:

- Alabama poultry segmentation.
- Agriculture-Vision.
- OpenEarthMap.
- One building-footprint dataset.

Fourth

Scale only the winning recipe to ViT-B and the full six-million-tile corpus.

Fifth

Run:

- Alabama full fine-tuning.
- Georgia zero-shot evaluation.
- Public benchmark evaluation.
- YOLO11 and YOLO26 application comparisons.

---

23. Feasible project boundary

Required for the main paper

- Exact overlap alignment.
- ViT-S ablations.
- ViT-B final model.

- Alabama and Georgia transfer.
- Agriculture-Vision.
- OpenEarthMap.
- Building segmentation.
- One oriented-detection dataset.
- Frozen, LoRA, and full fine-tuning.
- Three downstream seeds.
- Geographic transfer analysis.

Valuable but optional

- ViT-L.
- Full context mosaic encoder.
- Multispectral extension.
- More states.
- Temporal change detection.
- Road graph extraction.
- Foundation-to-YOLO distillation.
- Statewide Georgia inference.

The core paper should remain focused. The title, three primary contributions, and spatial diagnostic suite now form a coherent CVPR argument.